

SIMATSENGINEERING

ASSESSMENTTOOL1

COURSENAME:ArtificialIntelligence

COURSECODE:CSA17

COURSEOUTCOMECOVERED

C05:*DesignAI-basedsystemsusingPySpark,RDDoperations,andintelligentagentarchitectures,using scalable systems and integrate components in distributed AI applications. (BTL 5)*

AssessmentToolWeightage:50%

QUESTIONS: 5

AnnexureA

Duration:1Hour
TotalMarks:50

DesignTask

CodeofConduct:

I M. VAISHNAV (192524251) certifythatthissubmissionismyoriginalworkandthatIhaveadhered to the guidelines specified for this assessment.

Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.

Signatureofthestudent:

. VAISHNAV (192524251)

DesignTask

1. **HybridAIRecommendationEngine Design**

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed dataprocessing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. **SmartDisasterDetection System**

Develop a distributed AI system using PySpark and RDDoperationstodetectnaturaldisasters (such as floods or earthquakes) from multi-source data streams(satelliteimages,sensors,and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. **AI-BasedFraudDetectionFramework**

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. **AutonomousSupplyChainOptimizationSystem**

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory,logistics,demandforecasting).DescribehowRDDtransformationssupport large-scale data processing and how multiple intelligent agents interacttomakedecentralized decisions for cost minimization and efficiency improvement.

5. **PersonalizedLearningAnalyticsPlatform**

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	DataFlow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent(5)	Good (4)	Average(3)	Poor(2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed

DataFlow&Processing Pipeline	Clear,logical, and well-defined data flow	Mostly clear	Somegaps	Poorlydefined
Innovation& Creativity	Highly innovative, realistic,and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation &Documentation	Veryclear,neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poorpresentation

1. Hybrid AI Recommendation Engine Design

Introduction

A Hybrid AI Recommendation Engine combines Content-Based Filtering and Collaborative Filtering to provide accurate and personalized recommendations. PySpark RDDs enable scalable processing of massive user-item datasets.

System Architecture

Data Collection Layer

User profiles

Product/movie information

User ratings and reviews

Data Processing Layer

PySpark RDDs process large datasets.

Data cleaning and feature extraction.

Recommendation Layer

Content-Based Recommendation

Collaborative Filtering Recommendation

Intelligent Agent Layer

User Preference Agent

Feedback Analysis Agent

Recommendation Optimization Agent

PySpark RDD Operations

map() – Extract user and item features.

filter() – Remove invalid records.

groupByKey() – Group ratings by users.

reduceByKey() – Calculate average ratings.

join() – Combine user and item information.

Intelligent Agent Role

Monitor user behavior.

Learn user preferences.

Adapt recommendations based on feedback.

Improve recommendation accuracy in real time.

Data Flow

User Data → RDD Processing → Hybrid Recommendation Model → Intelligent Agents
→ Personalized Recommendations

Advantages

High recommendation accuracy.

Real-time adaptation.

Scalable distributed processing.

Better user satisfaction.

2. Smart Disaster Detection System

Introduction

A Smart Disaster Detection System uses distributed AI and PySpark to detect disasters such as floods, earthquakes, and cyclones using data from sensors, satellites, and social media.

System Architecture

Data Sources

Satellite images

IoT sensors

Social media feeds

Data Processing Layer

PySpark RDDs process incoming data streams.

Data cleaning and normalization.

AI Analytics Layer

Image classification models

Sensor anomaly detection

Social media sentiment analysis

Intelligent Agent Layer

Monitoring Agent

Alert Agent

Response Coordination Agent

Data Fusion Techniques

Sensor data fusion

Image and text fusion

Multi-source event correlation

PySpark RDD Operations

map()
flatMap()
filter()
reduceByKey()
Intelligent Agent Coordination
Detect disaster signals.
Verify information from multiple sources.
Issue warnings.
Coordinate emergency response.

Data Flow

Data Sources → RDD Processing → AI Analysis → Intelligent Agents → Early Warning System

Benefits

Faster disaster detection.
Improved public safety.
Real-time monitoring.
Scalable architecture.

3. AI-Based Fraud Detection Framework

Introduction

The system identifies fraudulent financial transactions using distributed anomaly detection algorithms implemented with PySpark RDDs.

System Architecture

Transaction Collection Layer
Data Preprocessing Layer
Fraud Detection Layer
Alert and Monitoring Layer
PySpark RDD Operations
map() – Extract transaction features.
filter() – Remove incomplete records.

reduceByKey() – Aggregate customer activity.

join() – Combine customer history.

Distributed Anomaly Detection

Unusual transaction amount.

Sudden location changes.

Abnormal spending patterns.

High-frequency transactions.

Intelligent Agents

Monitoring Agent

Tracks transactions continuously.

Detection Agent

Identifies suspicious activities.

Alert Agent

Sends notifications to banks and customers.

Learning Agent

Updates fraud patterns using new data.

Data Flow

Transactions → RDD Processing → Fraud Detection Model → Intelligent Agents
→ Fraud Alerts

Advantages

Real-time fraud detection.

Reduced financial losses.

Continuous learning.

Scalable big-data processing.

4. Autonomous Supply Chain Optimization System

Introduction

This AI-driven system optimizes inventory management, logistics, and demand forecasting using PySpark and intelligent agents.

System Architecture

Inventory Management Module

Logistics Management Module

Demand Forecasting Module

Agent Coordination Module

PySpark RDD Operations

map() – Process inventory data.

filter() – Remove invalid entries.

groupByKey() – Group products by category.

reduceByKey() – Calculate stock levels.

Intelligent Agents

Inventory Agent

Maintains stock levels.

Logistics Agent

Optimizes transportation routes.

Demand Forecast Agent

Predicts future product demand.

Decision Agent

Coordinates supply chain decisions.

Data Flow

Inventory Data → RDD Processing → Forecasting Models → Intelligent Agents
→ Optimized Supply Chain

Scalability Features

Distributed computing.

Fault tolerance.

Parallel processing.

Real-time decision making.

Benefits

Reduced operational costs.

Improved delivery efficiency.

Better inventory control.

Increased customer satisfaction.

5. Personalized Learning Analytics Platform

Introduction

This platform analyzes student learning behavior using AI and PySpark to provide adaptive learning experiences.

System Architecture

Student Data Collection Layer

Learning Analytics Layer

Personalization Engine

Intelligent Agent Layer

Data Sources

Quiz scores

Attendance records

Learning activities

Course completion data

PySpark RDD Operations

map() – Extract learning metrics.

filter() – Remove incomplete records.

reduceByKey() – Calculate performance scores.

join() – Combine academic records.

Intelligent Agents

Student Monitoring Agent

Tracks learning progress.

Feedback Agent

Provides personalized suggestions.

Recommendation Agent

Suggests learning materials.

Performance Agent

Identifies weak areas.

Data Flow

Student Activities → RDD Processing → Analytics Engine → Intelligent Agents
→ Personalized Content Delivery

Adaptive Learning Features

Personalized content.

Real-time feedback.

Learning path recommendations.

Performance prediction.

Benefits

Improved learning outcomes.

Better student engagement.

Scalable educational analytics.

Support for thousands of learners.